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Author Name	Madhur Suman
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Submitted Text

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2	34	5	2	0.236 %	view

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1	79	10	2	0.472 %	view
2	30	7	2	0.330 %	view
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4	40	6	2	0.283 %	view
5	36	6	2	0.283 %	view
6	39	6	2	0.283 %	view
7	32	6	2	0.283 %	view
8	35	5	2	0.236 %	view
9	42	5	2	0.236 %	view
10	25	5	2	0.236 %	view
11	29	5	2	0.236 %	view
12	39	5	2	0.236 %	view
13	31	5	2	0.236 %	view
14	34	4	2	0.188 %	view
15	39	4	2	0.188 %	view
16	19	4	2	0.188 %	view
17	30	4	2	0.188 %	view
18	30	4	2	0.188 %	view
19	26	4	2	0.188 %	view
20	28	4	2	0.188 %	view
21	21	4	2	0.188 %	view

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Sl. No	Index	Lines	Words	% in Report	
1	Abstract iv	4	12	0.283 %	view
2	1.1 Introduction 2	8	29	0.684 %	view
3	Chapter title page	16	19	0.448 %	view
4	Chapter 2	17	85	2.007 %	view
5	9 Chapter 1	95	1017	24.01 %	view
6	13 Chapter 2	34	271	6.399 %	view
7	15 Chapter 3	31	271	6.399 %	view
8	17 Chapter 4	150	1032	24.36 %	view
9	24 Chapter 5	40	278	6.564 %	view
10	5.2 Conclusion	70	573	13.53 %	view
11	Other Data	64	547	12.91 %	view

Submitted Text:

Line 1| 1 DEEPFAKE DETECTION SYSTEM
Line 2| A project submitted in partial fulfilment of the requirements
Line 3| [for the award of the degree of](#)
Line 4| Bachelor of Technology
Line 5| [in Computer Science and Engineering](#)
Line 6| Group Members:
Line 7| [Sankalp \(12311001\)](#)
Line 8| [Madhur Suman \(12311002\)](#)
Line 9| Supervised By:
Line 10| Dr. Md. Arquam
Line 11| (Assistant Professor)
Line 12| INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SONEPAT
Line 13| 131201, HARYANA, INDIA
Line 14| 2 ACKNOWLEDGEMENT None of this would have come together without the people who nudged us forward when the
Line 15| experiments stalled and the plots looked wrong. We owe a sincere note of thanks to everyone
Line 16| who pitched in.
Line 17| Dr. Md. Arquam, Assistant Professor in the CSE department at IIIT Sonapat, let us take on
Line 18| this topic and stayed reachable even when his calendar was packed. That steady backing
Line 19| mattered more than we can fit on one page.
Line 20| Other faculty in CSE also steered us toward better reading on [deep learning](#) and multimedia
Line 21| forensics. Their comments in reviews tightened both the code paths we chose and the way we
Line 22| wrote them up.
Line 23| Last, our families and friends put up with late nights and last-minute runs; we are grateful for
Line 24| that patience.
Line 25| Sankalp Madhur Suman
Line 26| 3 SELF DECLARATION
Line 27| We hereby state that the work contained in the project [titled "Deepfake Detection System"](#) is
Line 28| original. We have followed the standards of project ethics to the best of our abilities. We
Line 29| have acknowledged all the sources of knowledge that we have used in this project.
Line 30| [Sankalp \(12311001\)](#)
Line 31| [Madhur Suman \(12311002\)](#)
Line 32| [Department of Computer Science and Engineering](#)
Line 33| [Indian Institute of Information](#) Technology,
Line 34| Sonapat - 131201, Haryana, India

Line 35| 4 CERTIFICATE This is to certify that Mr. Sankalp and Mr. Madhur Suman have worked on the project
Line 36| entitled "Deepfake Detection System" under my supervision and guidance.
Line 37| The contents of the project, being submitted to the Department of [Computer Science and](#)
Line 38| [Engineering, IIIT Sonapat, Haryana, for the award of the degree of B.Tech in Computer](#)
Line 39| [Science and Engineering, are original and carried out by the candidates themselves. This](#)
Line 40| [project has not been submitted in full or part for the award of any other degree or diploma to](#)
Line 41| [this or any other university.](#)
Line 42| [Dr. Md. Arquam,](#)
Line 43| [Supervisor Department of Computer Science and Engineering.](#)
Line 44| [Indian Institute of Information](#) Technology,
Line 45| Sonapat, Haryana
Line 46| 5 ABSTRACT Project Title: Deepfake Detection System
Line 47| Submitted by: 1. Sankalp (12311001) 2. Madhur Suman (12311002)
Line 48| Degree: B.Tech
Line 49| Project Supervisor: Dr. Md. Arquam
Line 50| Month and Year: April, 2026
Line 51| Department: [Computer Science and Engineering, IIIT Sonapat, Haryana](#)
Line 52| We built a detector that listens as well as it looks. Synthetic clips often [break down](#) at the
Line 53| boundary between picture and sound—lip timing slips, the voice does not quite match the
Line 54| face, or a splice shows up only when both tracks are compared. A ViT-B/16 backbone reads
Line 55| sixteen sampled frames; a compact four-layer CNN turns mel spectrograms into tokens; a
Line 56| [Multimodal Bottleneck Transformer \(MBT\)](#) lets those streams talk without full cross-
Line 57| attention everywhere.
Line 58| Training is split into two stages. First, on unlabelled [VoxCeleb](#) video, we run three self-
Line 59| supervised losses together: MoCo-style cross-modal contrastive learning [with a memory](#)
Line 60| [bank](#), a sync-vs-desync classifier, and masked reconstruction for both modalities. Second, we
Line 61| load those encoders into a binary head and fine-tune on FakeAVCeleb. Everything runs in
Line 62| PyTorch with automatic mixed precision. On [FakeAVCeleb](#) validation we reached 85.2%
Line 63| accuracy, 86.42% precision, 75.6% recall, and an F1 of 80.1%.
Line 64| 6 TABLE OF CONTENTS
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Line 102| 8 LIST OF ABBREVIATIONS

Line 103| ABBREVIATION FULL FORM

Line 104| AI Artificial Intelligence

Line 105| ML Machine Learning

Line 106| DL Deep Learning

Line 107| CNN Convolutional Neural Network

Line 108| ViT Vision Transformer

Line 109| MBT Multimodal Bottleneck Transformer

Line 110| GAN Generative Adversarial Network

Line 111| SSL Self-Supervised Learning

Line 112| MoCo Momentum Contrast

Line 113| AV Audio-Visual

Line 114| GPU Graphics Processing Unit

Line 115| CPU Central Processing Unit

Line 116| API Application Programming Interface

Line 117| BCE Binary Cross-Entropy

Line 118| AMP Automatic Mixed Precision

Line 119| MLP Multi-Layer Perceptron

Line 120| GELU Gaussian Error Linear Unit

Line 121| [InfoNCE](#) Information Noise Contrastive Estimation

Line 122| FPS Frames Per Second

[Line 123](#)| 9 Chapter 1

Line 124| Introduction 1.1 INTRODUCTION

Line 125| Cheap generative models can now fabricate faces, voices, and short clips that look fine at
Line 126| first glance. People call the worst of these deepfakes—often built with GANs, diffusion nets,
Line 127| or heavy autoencoder stacks. A creator can paste a face, mimic a voice, or nudge expressions
Line 128| until the clip passes casual viewing. The same line of research that produced demos for
Line 129| papers has spilled into scams, impersonation, and coordinated disinformation, so trust in
Line 130| ordinary video is shakier than it was even five years ago.

Line 131| Manual inspection does not scale, and old fingerprint tricks (compression quirks, blockiness)
Line 132| miss many modern fakes because the synthesis is too clean. We still need detectors that run
Line 133| on GPUs people can rent, update when generators change, and flag content before it spreads.
Line 134| We lean on both modalities on purpose. Plenty of prior work stares at frames alone; we pair
Line 135| RGB sequences with audio because many forgeries show up only in the coupling—mouth
Line 136| shape lagging syllables, a laugh that arrives half a beat late, or speech spliced from another
Line 137| take. A multimodal model can chase those tells.

Line 138| Pretraining is label-free on real speech video (VoxCeleb), then we snap on a small supervised
Line 139| head for FakeAVCeleb. ViT-B/16 carries the visual side, a four-layer CNN maps mel bins to
Line 140| tokens, and MBT sits in the middle so the streams exchange information through shared
Line 141| bottlenecks instead of all-to-all attention.

Line 142| 1.2 PROBLEM OUTLINE

Line 143| The literature is large, yet a few pain points keep showing up in practice.

Line 144| Vision-only detectors chase spatial cues—blurred jawlines, odd pores, lighting that flickers
Line 145| between frames. That can work inside the benchmark they were tuned on, yet a new
Line 146| generator may hide those cues altogether, so pure frame classifiers age quickly.

Line 147| 10 Labels are scarce and expensive. Public packs cover a slice of methods and scenes, so a
Line 148| model that memorizes one corpus often stumbles outdoors the moment the attack recipe
Line 149| changes. Audio is frequently dropped even though many manipulations leave the soundtrack
Line 150| misaligned or borrowed from elsewhere. Ignoring the waveform throws away half the story.
Line 151| Video is heavy: sixteen frames per clip through a ViT is not lightweight, so any serious
Line 152| pipeline must trade resolution, batch size, and wall-clock time against what a lab or a startup
Line 153| can actually afford.

Line 154| 1.3 PROJECT OBJECTIVES

Line 155| We set out to ship a detector that is multimodal, partly self-supervised, measurable on
Line 156| FakeAVCeleb, and split into modules we can swap later.

Line 157| • Cross-modal cues. Fuse RGB tokens and mel tokens through MBT so timing and
Line 158| appearance are judged jointly rather than in isolation.
Line 159| • Self-supervised front end. Pretrain without fake labels on [VoxCeleb](#) using contrastive
Line 160| alignment, sync discrimination, and masked reconstruction, then fine-tune on
Line 161| [FakeAVCeleb](#) with a thin classifier head.

Line 162| • Benchmarks and budget. Report accuracy, precision, recall, and F1 while staying
Line 163| within reach of a mid-range GPU via AMP and sensible batching.

Line 164| • Clean seams. Keep encoders, fusion, and head in separate classes so we can drop in a
Line 165| stronger ViT or audio model without rewriting the whole repo.

Line 166| 1.4 PROJECT METHODOLOGY

Line 167| We moved in six concrete steps from papers to checkpoints.

Line 168| 1. Reading and stack pick. After skimming forgery surveys and multimodal fusion work, we
Line 169| settled on PyTorch, ViT-B/16 for frames, and MBT-style bottlenecks for fusion.

Line 170| 11 2. Data pulls. [VoxCeleb](#) feeds SSL; [FakeAVCeleb](#) supplies labelled real/fake pairs. Scripts
Line 171| sample sixteen frames, crop centres, apply ImageNet stats, and build log-mel spectrograms at
Line 172| 16 kHz.

Line 173| 3. Code layout. [VideoViTEncoder](#) wraps [torchvision](#) ViT with temporal position codes;
Line 174| [AudioTokenEncoder](#) is a four-layer CNN into 768-D; [AudioVisualFusionEngine](#) stacks
Line 175| MBT layers with four shared tokens.

Line 176| 4. SSL loop. Contrastive [InfoNCE with a memory bank](#), sync negatives (swapped clips and
Line 177| time shifts), and masked blocks on both modalities. [ssl_pretrain.py](#) is set to 50 epochs
Line 178| (AdamW, AMP, grad clip 1.0); Table 5.1 / Figure 5.1 show an 8-epoch snapshot from our
Line 179| logged run.

Line 180| 5. Fine-tune. [Load SSL weights into](#) DeepfakeClassifier, attach a three-layer MLP head, and
Line 181| use a lower LR on pretrained blocks than on the head.

Line 182| 6. Numbers. Track loss curves in TensorBoard, dump PNG dashboards for finetune metrics,
Line 183| and score validation with accuracy / precision / recall / F1.

Line 184| 1.5 SCOPE OF PROJECT WORK

Line 185| In scope: end-to-end training code, not a polished consumer app.

Line 186| • Dual encoders. Sixteen RGB frames and matching mel tensors, each mapped to 768-D
Line 187| sequences with temporal embeddings.

Line 188| • Fusion. Four MBT layers, four bottleneck tokens—information passes through the
Line 189| narrow slots instead of dense audio-visual attention.

Line 190| • Pretrain stack. SSL with flips, colour jitter, light blur, gain noise, [SpecAugment](#)
Line 191| masks, MoCo bank, sync games, and masked reconstruction.

Line 192| • Supervised wrap-up. Finetune with 80/20 split, BCE-with-logits, cosine schedule,
Line 193| AMP, and plots for loss and classification metrics.

Line 194| 12 1.6 LIMITATIONS

Line 195| • GPU memory. SSL with full ViT forward passes wants VRAM; 16 GB is a rough
Line 196| floor and 24 GB (3090/4090 class) is more comfortable.

Line 197| • Domain shift. [VoxCeleb](#) is interview-heavy; [FakeAVCeleb](#) reflects certain synthesis
Line 198| recipes—new attacks may still slip past.

Line 199| • Missing sound. No audio means our design loses half its signal; we do not ship a
Line 200| visual-only fallback path in code.

Line 201| • Latency. Training scripts batch offline clips; live moderation would need quantisation,
Line 202| distillation, or smaller backbones.

Line 203| We did not run targeted adversarial evasion tests; a patient attacker could optimise against
Line 204| our logits, which is left for future work.
Line 205| 1.7 ORGANIZATION OF THE PROJECT
Line 206| Five chapters walk from motivation to numbers.
Line 207| • Chapter 1 Motivation, goals, how we worked, what we included, and where we
Line 208| stopped. • Chapter 2 Python stack and the CLI workflow for download → preprocess → train.
Line 209| • Chapter 3 Stages from MP4 on disk to [logits out of the](#) classifier.
Line 210| • Chapter 4 Encoders, fusion, losses, loading checkpoints, and augmentation—with
Line 211| code fragments.
Line 212| • Chapter 5 Curves, tables, takeaway, and what we would try next.

Line 213| 1.8 SUMMARY

Line 214| Chapter 1 set the scene: forgeries are easier to make, harder to eyeball, and often visible only
Line 215| when audio and video disagree. We listed what blocks deployment—narrow vision cues,
Line 216| label shortage, ignored waveforms, compute—and described how we train in two phases with
Line 217| MBT in the middle. Later chapters unpack scripts, tensors, and metrics.

Line 218| 13 [Chapter 2](#)

Line 219| [Application and User Interface](#)

Line 220| 2.1 TECHNOLOGIES EMPLOYED

Line 221| Everything is Python-first; the list below is the glue we actually import day to day.
Line 222| • Runtime. Python 3.8+ for models, dataloaders, and glue scripts.
Line 223| • Core DL. PyTorch 2.x modules (nn, optim, amp) for forward/backward and CUDA
Line 224| kernels. • Vision. TorchVision: ViT-B/16 weights plus read_video and the usual
Line 225| resize/crop/normalise helpers.
Line 226| • Audio. TorchAudio: resample to 16 kHz, MelSpectrogram, AmplitudeToDB.
Line 227| • Plots. Matplotlib for PNG loss boards after each phase.
Line 228| • Live scalars. [TensorBoard](#) while SSL runs long.
Line 229| • Progress. [tqdm](#) on every epoch loop so we see ETA when a run stretches overnight.
Line 230| • Downloads. [gdown](#) and requests for scripted dataset fetches.

Line 231| [2.2 USER INTERFACE DESCRIPTION](#)

Line 232| There is no GUI—only terminals, flags in code, and saved weights. That keeps the footprint
Line 233| small for a coursework repo.

Line 234| Training Pipeline Interface

Line 235| Rough order of operations:

- Line 236| 1. [dataset/pretrain_datset_download.py](#) — pull VoxCeleb, unpack archives.
- Line 237| 2. [dataset/pretrain_prepare_dataset.py](#) — flatten folders so loaders see a simple list of MP4
Line 238| paths. 14 3. [dataset/preprocess_dataset.py](#) — bake frames + mel tensors to .pt shards to avoid decoding
Line 239| every epoch.
- Line 240| 4. [ssl_pretrain.py](#) — run SSL objectives, stream scalars to TensorBoard, stash checkpoints
Line 241| under outputs/.
- Line 242| 5. [dataset/prepare_finetune_dataset.py](#) — park [FakeAVCeleb](#) clips under [real/](#) and [fake/](#) with
Line 243| balanced counts.
- Line 244| 6. [finetune_deepfake.py](#) — load SSL weights, train the binary head, keep the best F1
Line 245| checkpoint, write metric PNGs.

Line 246| Inference Interface

Line 247| At test time you torch.load the best checkpoint, push tensors through the same preprocessors,
Line 248| apply sigmoid to the logit, and treat scores above 0.5 as fake.

Line 249| Output Artifacts

Line 250| Expect loss PNGs, [TensorBoard](#) event files, finetune dashboards under metrics/, and .pth
Line 251| checkpoints you can copy elsewhere.

Line 252| 15 [Chapter 3](#)

Line 253| [System and Pipeline Design](#)

Line 254| [3.1 INTRODUCTION TO THE SYSTEM PIPELINE](#)

Line 255| A clean pipeline saves debugging time: you can unit-test preprocessing without touching
Line 256| fusion, and swap encoders without rewriting dataloaders. Multimodal work doubles the
Line 257| moving parts—waveform and pixels must stay time-aligned.
Line 258| Data enters as MP4, splits into RGB and audio tensors, passes through separate encoders,

Line 259] meets inside MBT, then lands in a shallow MLP that emits one logit.

Line 260] [3.2 PROJECT SYSTEM DESCRIPTION](#)

Line 261] Five stages, mapped roughly to folders in the repo.

Line 262] Stage 1: Data Ingestion and Splitting

Line 263] [DeepfakeDataset](#) walks [real/](#) and [fake/](#), stores (path, label), shuffles, and holds an 80/20 split.

Line 264] [DataLoader](#) workers pin memory so batches reach the GPU without stalling the CPU.

Line 265] Stage 2: Preprocessing

Line 266] Video and audio split here.

Line 267] RGB: [linspace](#) sixteen indices across the clip, centre-crop to 0.8 of the short side, resize to

Line 268] 224^2 , normalise with ImageNet μ/σ .

Line 269] Audio: mono downmix, 16 [kHz](#) resample, 1024 FFT $\searrow$ 512 hop mel with 128 bins,

Line 270] AmplitudeToDB, pad/trim to ninety-four time bins so batches stack cleanly.

Line 271] Stage 3: Feature Encoding

Line 272] [VideoViTEncoder](#) runs ViT-B/16 per frame, keeps CLS tokens, adds learned temporal

Line 273] positions. 16 [AudioTokenEncoder](#) stacks four conv blocks (frequency shrinks, time mostly preserved),

Line 274] projects to 768-D, adds temporal positions.

Line 275] Stage 4: Multimodal Fusion

Line 276] [AudioVisualFusionEngine](#) prepends CLS tokens, seeds four bottleneck vectors, and stacks

Line 277] four MBT layers. Each modality self-attends with bottlenecks—not with the other stream

Line 278] directly—then bottlenecks are averaged across modalities before the next layer.

Line 279] Stage 5: Classification

Line 280] Concatenate final visual and audio CLS (1536-D), MLP $1536 \rightarrow 512 \rightarrow 256 \rightarrow 1$ with GELU +

Line 281] dropout, sigmoid at inference with a 0.5 cut.

Line 282] AMP wraps training steps; gradients clip at norm 1.0 to stop spikes.

Line 283] [17 Chapter 4](#)

Line 284] [Model Design and Logic](#)

Line 285] [4.1 DEEP LEARNING MODELS USED](#)

Line 286] Three trainable pieces: a frame encoder, a spectrogram encoder, and the MBT stack that

Line 287] negotiates between them.

Line 288] Vision Transformer (ViT-B/16) as Visual Encoder

Line 289] ViT-B/16 chops each 224^2 frame into 16×16 patches, embeds them, and stacks transformer

Line 290] blocks—see Dosovitskiy et al. for the full recipe.

Line 291] We start from IMAGENET1K_V1 weights, strip the classifier head for an identity, and emit

Line 292] one 768-D CLS vector per frame. [VideoViTEncoder](#) adds temporal position codes across the

Line 293] sixteen CLS outputs so fusion sees order, not just appearance.

Line 294] CNN-Based Audio Tokenizer as Audio Encoder

Line 295] Mel maps stay 2-D; the CNN squeezes frequency while leaving time mostly untouched until

Line 296] late pooling, yielding a token per time step.

Line 297] Four 3×3 conv blocks use [GroupNorm](#) + ReLU; early max-pool steps shrink freq and time,

Line 298] later (2,1) pools attack frequency only. Adaptive pooling collapses the last freq axis; a linear

Line 299] + [LayerNorm](#) maps $256 \rightarrow 768$, then temporal positions match the video side.

Line 300] [Multimodal Bottleneck Transformer \(MBT\)](#) as Fusion Engine

Line 301] Following Nagrani et al., audio and vision never attend across streams wholesale—they read

Line 302] and write four shared bottleneck tokens instead.

Line 303] Four MBT layers, each with parallel ViT-style blocks per modality, eight heads, dropout 0.1:

Line 304] bottlenecks append to both sequences, attention runs, bottlenecks merge by averaging, repeat.

Line 305] 18 4.2 MODEL LOADING

Line 306] Weights arrive twice: ImageNet for ViT at construction, later SSL checkpoints when the

Line 307] classifier spins up.

Line 308] Pretrained ViT Loading

Line 309] [VideoViTEncoder](#) pulls [torchvision](#) weights like this:

Line 310] weights = [models.ViT_B_16_Weights.IMAGENET1K_V1](#) if pretrained else None

Line 311] self.vit = [models.vit_b_16\(weights=weights\)](#)

Line 312] self.vit.heads = nn.Sequential(nn.Identity())

Line 313] Spatial filters start strong; temporal embeddings draw fresh noise ($\sigma \approx 0.02$) because SSL must

Line 314] learn order from video.

Line 315| SSL Encoder Loading for Fine-Tuning

Line 316| [DeepfakeClassifier](#) accepts either bundled encoder dicts or prefixed state keys—handy when

Line 317| a checkpoint comes from the full SSL module.

Line 318| if [os.path.exists\(pretrained_path\)](#):

Line 319| checkpoint = [torch.load\(pretrained_path, map_location='cpu'\)](#)

Line 320| if 'visual' in checkpoint:

Line 321| [self.visual_encoder.load_state_dict\(checkpoint\['visual'\], strict=False\)](#)

Line 322| [self.audio_encoder.load_state_dict\(checkpoint\['audio'\], strict=False\)](#)

Line 323| [self.fusion_engine.load_state_dict\(checkpoint\['fusion'\], strict=False\)](#)

Line 324| else:

Line 325| visual_state = [{k.replace\('visual_encoder.', ''\): v](#)

Line 326| for k, v in checkpoint.items()

Line 327| if [k.startswith\('visual_encoder.'\)}](#)

Line 328| strict=False tolerates small name shifts between training scripts; missing keys simply stay

Line 329| random. After [ssl_pretrain.py](#) finishes, encoders are saved as [outputs/ssl_encoders_fused.pth](#)

Line 330| (visual/audio/fusion keys); [finetune_deepfake.py](#) points there by default.

Line 331| 4.3 PREDICTION LOGIC

Line 332| Forward pass is short: encode, fuse, concat CLS, classify. Reference implementation:

Line 333| def forward(self, visual, audio):

Line 334| v_tokens = [self.visual_encoder\(visual\)](#)

Line 335| a_tokens = [self.audio_encoder\(audio\)](#)

Line 336| fusion_out = [self.fusion_engine\(v_tokens, a_tokens\)](#)

Line 337| vis_cls = fusion_out["vis_cls"]

Line 338| aud_cls = fusion_out["aud_cls"]

Line 339| combined = [torch.cat\(\[vis_cls, aud_cls\], dim=1\)](#)

Line 340| logits = self.classifier(combined)

Line 341| return logits

Line 342| Inputs: B×T×3×224×224 video and matching mel tensor. ViT emits sixteen 768-D tokens;

Line 343| CNN emits a time sequence at 768-D.

Line 344| MBT returns fused CLS summaries per modality after four layers; we concatenate for a

Line 345| 1536-D joint vector.

Line 346| Head: GELU MLP, BCE-with-logits in training, sigmoid + 0.5 threshold when evaluating.

Line 347| 4.4 DATA PROCESSING

Line 348| [VisualPreprocessor](#) and [AudioPreprocessor](#) share the same math in SSL and fine-tune so

Line 349| tensors [line up](#) with whatever the encoders expect.

Line 350| Visual Preprocessing

Line 351| Frame sampling uses [torch.linspace](#) across the decoded stack:

Line 352| def [_sample_frames\(self, frames\)](#):

Line 353| total_frames = frames.shape[0]

Line 354| if total_frames == [self.num_frames](#):

Line 355| return frames

Line 356| indices = [torch.linspace\(0, total_frames - 1, self.num_frames\).long\(\)](#)

Line 357| return frames[indices]

Line 358| Then centre-crop, resize, ImageNet normalise—nothing exotic, just consistent.

Line 359| Audio Preprocessing

Line 360| Mel + dB scaling; key [hyperparameters](#) below.

Line 361| 20 [self.mel_transform](#) = [torchaudio.transforms.MelSpectrogram](#)(

Line 362| sample_rate=sample_rate, n_fft=n_fft,

Line 363| hop_length=hop_length, n_mels=n_mels, normalized=True

Line 364|) [self.amplitude_to_db](#) = [torchaudio.transforms.AmplitudeToDB](#)(

Line 365| stype='power', top_db=80

Line 366| Log-mel matches human loudness perception and tends to exaggerate [narrowband](#) glitches.

Line 367| Fixed ninety-four time bins keep batches square.

Line 368| Batch Preprocessing for SSL

Line 369| [preprocess_dataset.py](#) bakes each clip once to disk; [SSLDataset](#) then mmap-reads tensors,

Line 370| which beats decoding MP4 every epoch.

Line 371 [4.5 TRAINING AND INFERENCE](#)

Line 372 SSL teaches alignment without labels; supervised fine-tune pins the decision boundary on
Line 373 FakeAVCeleb. Phase 1: Self-Supervised Pretraining
Line 374 Three losses run jointly on VoxCeleb.
Line 375 Cross-modal contrastive: project fused CLS tokens to 256-D L2-normalised space, [InfoNCE](#)
Line 376 with in-batch negatives plus a 2048-slot MoCo bank (momentum 0.999) and a light variance
Line 377 term so embeddings do not collapse.
Line 378 Sync task: half the [minibatch](#) stays aligned; half is corrupted—30% with foreign audio, 70%
Line 379 with time-shifted audio (15–40 frames). The head learns aligned vs not.
Line 380 Masked reconstruction: block masks (visual ~85% in 8-token chunks, audio ~80% in 12-bin
Line 381 chunks), smooth L1 only on masked units.
Line 382 Loss weights: 0.5 contrastive, 1.0 sync, 2.0 audio recon, 3.0 visual recon, 0.1 variance. Fifty
Line 383 epochs, [AdamW](#) 1e-4, wd 1e-4, four-step accumulation (effective batch 32), AMP, grad clip
Line 384 21 Phase 2: Supervised Fine-Tuning
Line 385 [Load SSL weights into](#) DeepfakeClassifier; split parameter groups so pretrained layers move
Line 386 slower:
Line 387 optimizer = torch.optim.AdamW([
Line 388 {"params": pretrained_params, "lr": LEARNING_RATE * 0.1},
Line 389 {"params": classifier_params, "lr": LEARNING_RATE}
Line 390], weight_decay=1e-4)
Line 391 Base LR 1e-4 on the head, 1e-5 on encoders/fusion. [finetune_deepfake.py](#) uses 20 epochs,
Line 392 batch size 4, BCE-with-logits, cosine decay, AMP, grad clip 1.0, best-F1 checkpoint kept as
Line 393 outputs_finetune/best_deepfake_detector.pth. Inference eval(), identical preprocessing, no gradient:
Line 394 checkpoint = [torch.load\("outputs_finetune/best_deepfake_detector.pth"\)](#)
Line 395 model = DeepfakeClassifier()
Line 396 [model.load_state_dict\(checkpoint\['model_state_dict'\]\)](#) model.eval()
Line 397 with [torch.no_grad\(\)](#):
Line 398 logits = model(video_tensor, audio_tensor)
Line 399 probability = torch.sigmoid(logits)
Line 400 is_fake = probability > 0.5
Line 401 Probabilities near one mean aggressive fake estimates; near zero the model leans real.

Line 402 [4.6 ERROR HANDLING](#)

Line 403 Broken clips should not kill a week-long run.

Line 404 Data Loading Errors

Line 405 [DeepfakeDataset](#) wraps [torchvision.io.read_video](#) in try/except; failures return zero tensors
Line 406 with the right shape so the iterator keeps moving (not ideal for learning, but stable).

Line 407 22 try: video, audio, info = [torchvision.io.read_video\(video_path, pts_unit='sec'\)](#)

Line 408 except Exception as e:

Line 409 print(f"Error loading video {video_path}: {e}")

Line 410 return [torch.zeros\(self.num_frames, 3, self.target_size, self.target_size\)](#)

Line 411 Silent audio falls back similarly—better than aborting mid-epoch.

Line 412 Dataset Validation

Line 413 If [real/](#) or [fake/](#) is empty, [__init__](#) raises immediately with the folder path in the message.

Line 414 if len(self.samples) == 0:

Line 415 raise RuntimeError(f"No videos found in {root_dir}/real or {root_dir}/fake")

Line 416 Model State Validation

Line 417 [VideoViTEncoder](#) asserts a five-dimensional input (B,T,C,H,W) before ViT runs.

Line 418 assert frames.dim() == 5, [f'Expected](#) 5D input (B, T, C, H, W), got {frames.dim()}D"

Line 419 Training Stability

Line 420 Grad clipping, AMP scaler, and the contrastive variance term together curb blow-ups and

Line 421 collapsed embeddings.

Line 422 [4.7 ADDITIONAL LOGIC: DATA AUGMENTATION](#)

Line 423 SSL augments aggressively; fine-tune sticks to clean tensors so distribution shift stays

Line 424 manageable. Video Augmentation

Line 425 VideoAugmentor: • 50% mirror flip, same flag for all frames in the clip.

Line 426 • 80% colour jitter (brightness/contrast/saturation 0.6–1.4), shared across frames.

Line 427 23 • 10% light Gaussian blur on every frame to mimic soft focus or heavy compression.

Line 428 Audio Augmentation

Line 429 AudioAugmentor: • Random gain 0.7–1.4 half the time.

Line 430 • Two time masks (≤ 10 bins each).

Line 431 • Two frequency masks (≤ 15 mel channels each).

Line 432 Fine-tune and inference skip these transforms.

Line 433 24 [Chapter 5](#)

Line 434 [Result and Conclusion](#)

Line 435 [5.1 RESULTS AND OBSERVATIONS](#)

Line 436 We watched SSL losses fall, then checked [FakeAVCeleb](#) metrics after fine-tune—both stages
Line 437 tell part of the story.

Line 438 Self-Supervised Pretraining Results

Line 439 SSL: the script trains up to 50 epochs; the table below matches the eight-epoch checkpoint

Line 440 we used for Figure 5.1 (same run as the loss snapshot in this report). [TensorBoard](#) has the full
Line 441 series if you train longer.

Line 442 Metric Initial Value Final Value (Epoch 8)

Line 443 Total Loss 1.50 0.60

Line 444 Contrastive Loss 1.40 0.04

Line 445 Sync Loss 0.24 0.05

Line 446 Audio Reconstruction Loss 0.19 0.20

Line 447 Visual Reconstruction Loss 0.06 0.05

Line 448 Total loss 1.50→0.60; contrastive 1.40→0.04 (pairs [line up](#) in embedding space); sync

Line 449 0.24→0.05; reconstructions hover low—so masking still teaches detail. Figure 5.1 overlays
Line 450 the five curves.

Line 451 Figure 5.1: Multimodal SSL training progress — per-objective losses (RunMetric snapshot or

Line 452 [outputs/loss_curve.png](#) from [ssl_pretrain.py](#))

Line 453 Fine-Tuning Results

Line 454 [FakeAVCeleb](#) validation scores:

Line 455 Metric Score

Line 456 Accuracy 85.2%

Line 457 Precision 86.42%

Line 458 Recall 75.6%

Line 459 F1 Score 80.1%

Line 460 85.2% accuracy is solid for the split we used. Precision 86.42% means flagged fakes are

Line 461 usually right; recall 75.6% leaves room—about one in four fakes slips through at the default

Line 462 threshold. F1 80.1% sits between those extremes.

Line 463 Precision beats recall: the model prefers marking doubtful clips as real, which tracks with a

Line 464 conservative 0.5 cut and class balance. Operators can move the threshold if false negatives

Line 465 hurt more than false positives.

Line 466 Figure 5.2 stitches loss, accuracy, and PR-F1 across the 20 fine-tuning epochs defined in

Line 467 [finetune_deepfake.py](#). 26

Line 468 Figure 5.2: Fine-tuning metrics dashboard — loss, accuracy, precision/recall/F1

Line 469 ([metrics/training_metrics_dashboard.png](#) from [finetune_deepfake.py](#), 20 epochs)

Line 470 Key Observations

Line 471 Audio+video fusion buys signal we would drop with vision-only SSL. Pretraining buys faster

Line 472 finetune convergence; freezing LR ratios stops catastrophic forgetting in the encoders.

Line 473 5.2 CONCLUSION

Line 474 Pairing ViT features, CNN mel tokens, and MBT fusion yielded 85.2% accuracy and 80.1%

Line 475 F1 on [FakeAVCeleb](#) under our split—enough to show the idea carries weight.

Line 476 SSL on real-only [VoxCeleb](#) cuts the labelled burden: the encoders already [know how](#) speech

Line 477 lines up with faces before fake labels appear.

Line 478 Modules are split on purpose—swap ViT for a newer variant, drop in an audio transformer,

Line 479 or add side channels without rewriting the repo from scratch.

Line 480 Bottleneck fusion is not the last word, but it is a workable baseline we can iterate from.

Line 481 5.3 FUTURE SCOPE

Line 482 Short list of what we would tackle next.

Line 483 Ship a thin FastAPI/Flask wrapper so reviewers can upload MP4s instead of editing paths in
Line 484 Python. Chase real-time: quantise, distil, or shrink backbones so a stream could be scored near frame
Line 485 rate. 27 Add optical flow, landmarks, or FFT cues as extra inputs—the file layout already invites new
Line 486 tensors. Benchmark beyond [FakeAVCeleb](#) (FaceForensics++, Celeb-DF, DFDC) and study domain
Line 487 shift explicitly.
Line 488 Explainability (attention rollout, saliency) plus on-device compression matter if anyone
Line 489 deploys this outside a lab.

Line 490 28 REFERENCES [1] Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. (2021). An Image is Worth 16x16 Words:
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Line 507 World Dataset for Deepfake Detection. In Proceedings of ACM MM (pp. 2382-2390).

Line 508 29 APPENDIX Project Repository
Line 509 GitHub: <https://github.com/yourusername/DeepDetect2>
Line 510 Project Structure
Line 511 [DeepDetect2/](#) [+-- Models/](#)
Line 512 | [+-- Visual_encoder.py](#) # ViT-based video encoder
Line 513 | [+-- audio_encoder.py](#) # 4-layer CNN audio [tokenizer](#)
Line 514 | [+-- Fusion_Engine.py](#) # Multimodal Bottleneck Transformer
Line 515 | [+-- ssl_tasks.py](#) # SSL pretraining orchestrator
Line 516 [+-- utils/](#)
Line 517 | [+-- visual_preprocessing.py](#) # Frame sampling, cropping, normalization
Line 518 | [+-- audio_preprocessing.py](#) # Mel spectrogram extraction
Line 519 | [+-- augmentations.py](#) # Video and audio augmentation
Line 520 [+-- dataset/](#)
Line 521 | [+-- pretrain_datset_download.py](#)
Line 522 | [+-- pretrain_prepare_dataset.py](#)
Line 523 | [+-- preprocess_dataset.py](#)
Line 524 | [+-- download_finetune_dataset.py](#)
Line 525 | [+-- prepare_finetune_dataset.py](#)
Line 526 [+-- ssl_pretrain.py](#) # Self-supervised pretraining (Phase 1)
Line 527 [+-- finetune_deepfake.py](#) # Supervised fine-tuning (Phase 2)
Line 528 [+-- outputs/](#) # [ssl_encoders_fused.pth](#), [loss_curve.png](#), per-
Line 529 epoch checkpoints
Line 530 [+-- outputs_finetune/](#) # Fine-tuned checkpoints
Line 531 [+-- metrics/](#) # Training metric plots
Line 532 [+-- README.md](#)
Line 533 Key Dependencies
Line 534 Python >= 3.8
Line 535 PyTorch >= 2.0
Line 536 [TorchVision](#) [TorchAudio](#) [tqdm](#) matplotlib [tensorboard](#) [gdown](#) requests
Line 537 30 Hardware Used for Training
Line 538 Component Specification

Line 539| GPU NVIDIA RTX 3090 (24 GB VRAM)

Line 540| RAM 32 GB DDR4

Line 541| Storage 500 GB SSD

Line 542| OS Ubuntu 22.04 LTS

*** End of submitted text ***

4. Grammar Info.

[Mistake and Suggestion details:](#)

Part of the sentence to be reviewed: '[deep learning](#)', at the line number 20

Category: **Compound Words**

Suggestion: Based on the context of the sentence, we recommend you to replace '[deep learning](#)' to '[deep-learning](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[break down](#)', at the line number 52

Category: **Compound Words**

Suggestion: Based on the context of the sentence, we recommend you to replace '[break down](#)' to '[breakdown](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[VoxCeleb](#)', at the line number 58

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[VoxCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[FakeAVCeleb](#)', at the line number 62

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[FakeAVCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[InfoNCE](#)', at the line number 121

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[InfoNCE](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[VoxCeleb](#)', at the line number 159

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[VoxCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[FakeAVCeleb](#)', at the line number 161

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[FakeAVCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[VoxCeleb](#)', at the line number 170

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[VoxCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['FakeAVCeleb'](#), at the line number 170

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'FakeAVCeleb' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['VideoViTEncoder'](#), at the line number 173

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'VideoViTEncoder' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['torchvision'](#), at the line number 173

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'torchvision' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['AudioTokenEncoder'](#), at the line number 174

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'AudioTokenEncoder' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['AudioVisualFusionEngine'](#), at the line number 174

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'AudioVisualFusionEngine' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['InfoNCE'](#), at the line number 176

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'InfoNCE' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['ssl_pretrain.py'](#), at the line number 177

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'ssl_pretrain.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['/'](#), at the line number 178

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '/' to 'Proper Slash Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: ['/'](#), at the line number 183

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '/' to 'Proper Slash Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: ['/'](#), at the line number 183

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '/' to 'Proper Slash Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: ['/](#), at the line number 183

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '/' to 'Proper Slash Mark.'

Example: [view](#)

Part of the sentence to be reviewed: ['SpecAugment'](#), at the line number 190

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'SpecAugment' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['VoxCeleb'](#), at the line number 197

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'VoxCeleb' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['FakeAVCeleb'](#), at the line number 197

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'FakeAVCeleb' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['logits'](#), at the line number 209

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'logits' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['out of the'](#), at the line number 209

Category: **Compound Words**

Suggestion: Based on the context of the sentence, we recommend you to replace 'out of the' to 'out-of-the'.

Example: [view](#)

Part of the sentence to be reviewed: ['TensorBoard'](#), at the line number 228

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'TensorBoard' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['tqdm'](#), at the line number 229

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'tqdm' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['gdown'](#), at the line number 230

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'gdown' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['dataset/pretrain_datset_download.py'](#), at the line number 236

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'dataset/pretrain_datset_download.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[dataset/pretrain_prepare_dataset.py](#)', at the line number 237

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[dataset/pretrain_prepare_dataset.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[dataset/preprocess_dataset.py](#)', at the line number 238

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[dataset/preprocess_dataset.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[ssl_pretrain.py](#)', at the line number 240

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[ssl_pretrain.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[dataset/prepare_finetune_dataset.py](#)', at the line number 242

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[dataset/prepare_finetune_dataset.py](#)' to 'Proper Slash Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: '[FakeAVCeleb](#)', at the line number 242

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[FakeAVCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[real](#)', at the line number 242

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[real](#)' to 'Proper Slash Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: '[fake](#)', at the line number 242

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[fake](#)' to 'Proper Slash Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: '[finetune_deepfake.py](#)', at the line number 244

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[finetune_deepfake.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[TensorBoard](#)', at the line number 250

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[TensorBoard](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[DeepfakeDataset](#)', at the line number 263

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[DeepfakeDataset](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[real/](#)', at the line number 263

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[real/](#)' to '[Proper Slash Mark.](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[DataLoader](#)', at the line number 264

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[DataLoader](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[linspace](#)', at the line number 267

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[linspace](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[kHz](#)', at the line number 269

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[kHz](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[/](#)', at the line number 269

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[/](#)' to '[Proper Slash Mark.](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[VideoViTEncoder](#)', at the line number 272

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[VideoViTEncoder](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[AudioTokenEncoder](#)', at the line number 273

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[AudioTokenEncoder](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[AudioVisualFusionEngine](#)', at the line number 276

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[AudioVisualFusionEngine](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[prepends](#)', at the line number 276

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[prepends](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[VideoViTEncoder](#)', at the line number 292

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[VideoViTEncoder](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[GroupNorm](#)', at the line number 297

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[GroupNorm](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[LayerNorm](#)', at the line number 299

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[LayerNorm](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[VideoViTEncoder](#)', at the line number 309

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[VideoViTEncoder](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[torchvision](#)', at the line number 309

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[torchvision](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[models.ViT_B_16_Weights.IMAGENET1K_V1](#)', at the line number 310

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[models.ViT_B_16_Weights.IMAGENET1K_V1](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[models.vit_b_16\(weights=weights\)](#)', at the line number 311

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[models.vit_b_16\(weights=weights\)](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[DeepfakeClassifier](#)', at the line number 316

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[DeepfakeClassifier](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[os.path.exists\(pretrained_path\):](#)', at the line number 318

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[os.path.exists\(pretrained_path\):](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[torch.load\(pretrained_path,](#)' at the line number 319

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[torch.load\(pretrained_path,](#)' to 'Proper Apostrophes Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[map_location='cpu'](#)', at the line number 319

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[map_location='cpu'](#)' to 'Proper Apostrophes Mark'.

Example: [view](#)

Part of the sentence to be reviewed: `'self.visual_encoder.load_state_dict(checkpoint['visual']).'` at the line number 321

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'self.visual_encoder.load_state_dict(checkpoint['visual']).'` to `'Proper Fullstop/Period'`.

Example: [view](#)

Part of the sentence to be reviewed: `'self.audio_encoder.load_state_dict(checkpoint['audio']).'` at the line number 322

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'self.audio_encoder.load_state_dict(checkpoint['audio']).'` to `'Proper Fullstop/Period'`.

Example: [view](#)

Part of the sentence to be reviewed: `'self.fusion_engine.load_state_dict(checkpoint['fusion']).'` at the line number 323

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'self.fusion_engine.load_state_dict(checkpoint['fusion']).'` to `'Proper Fullstop/Period'`.

Example: [view](#)

Part of the sentence to be reviewed: `'{k.replace('visual_encoder.':'` at the line number 325

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'{k.replace('visual_encoder.':'` to `'Proper Apostrophes Mark.'`

Example: [view](#)

Part of the sentence to be reviewed: `'")):'` at the line number 325

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'')):'` to `'Proper Apostrophes Mark.'`

Example: [view](#)

Part of the sentence to be reviewed: `'k.startswith('visual_encoder.'))'` at the line number 327

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'k.startswith('visual_encoder.'))'` to `'Proper Fullstop/Period'`.

Example: [view](#)

Part of the sentence to be reviewed: `'ssl_pretrain.py.'` at the line number 329

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'ssl_pretrain.py'` to `'Proper Fullstop/Period'`.

Example: [view](#)

Part of the sentence to be reviewed: `'outputs/ssl_encoders_fused.pth'` at the line number 329

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'outputs/ssl_encoders_fused.pth'` to `'Proper Fullstop/Period'`.

Example: [view](#)

Part of the sentence to be reviewed: `'finetune_deepfake.py'` at the line number 330

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace `'finetune_deepfake.py'` to `'Proper Fullstop/Period'`.

Example: [view](#)

Part of the sentence to be reviewed: '[self.visual_encoder\(visual\)](#)', at the line number 334

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[self.visual_encoder\(visual\)](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[self.audio_encoder\(audio\)](#)', at the line number 335

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[self.audio_encoder\(audio\)](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[self.fusion_engine\(v_tokens,](#)' at the line number 336

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[self.fusion_engine\(v_tokens,](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[torch.cat\(\[vis_cls,](#)' at the line number 339

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[torch.cat\(\[vis_cls,](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[logits](#)', at the line number 340

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[logits](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[logits](#)', at the line number 341

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[logits](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[VisualPreprocessor](#)', at the line number 348

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[VisualPreprocessor](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[AudioPreprocessor](#)', at the line number 348

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[AudioPreprocessor](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[line up](#)', at the line number 349

Category: **Compound Words**

Suggestion: Based on the context of the sentence, we recommend you to replace '[line up](#)' to 'lineup'.

Example: [view](#)

Part of the sentence to be reviewed: '[_sample_frames\(self,](#)' at the line number 352

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[_sample_frames\(self,](#)' to 'Proper Hyphens Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: '[self.num_frames](#):', at the line number 354

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[self.num_frames](#):' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[self.num_frames](#)).long()', at the line number 356

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[self.num_frames](#)).long()' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[hyperparameters](#)', at the line number 360

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[hyperparameters](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[self.mel_transform](#)', at the line number 361

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[self.mel_transform](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[self.amplitude_to_db](#)', at the line number 364

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[self.amplitude_to_db](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[narrowband](#)', at the line number 366

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[narrowband](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[preprocess_dataset.py](#)', at the line number 369

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[preprocess_dataset.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[SSLDataset](#)', at the line number 369

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[SSLDataset](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[InfoNCE](#)', at the line number 375

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[InfoNCE](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[minibatch](#)', at the line number 378

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[minibatch](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[AdamW](#)', at the line number 383

Category: **Orthographic Error**

Suggestion: Based on the context of the sentence, we recommend you to replace '[AdamW](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[{"params":](#)', at the line number 388

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[{"params":](#)' to 'Proper Quotation Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[{"params":](#)', at the line number 389

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[{"params":](#)' to 'Proper Quotation Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[finetune_deepfake.py](#)', at the line number 391

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[finetune_deepfake.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[torch.load\("outputs_finetune/best_deepfake_detector.pth"\)](#)', at the line number 394

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[torch.load\("outputs_finetune/best_deepfake_detector.pth"\)](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[model.load_state_dict\(checkpoint\["model_state_dict"\]\)](#)', at the line number 396

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[model.load_state_dict\(checkpoint\["model_state_dict"\]\)](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[torch.no_grad\(\):](#)', at the line number 397

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[torch.no_grad\(\):](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[logits](#)', at the line number 398

Category: **Orthographic Error**

Suggestion: Based on the context of the sentence, we recommend you to replace '[logits](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[DeepfakeDataset](#)', at the line number 405

Category: **Orthographic Error**

Suggestion: Based on the context of the sentence, we recommend you to replace '[DeepfakeDataset](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[torchvision.io.read_video](#)', at the line number 405

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'torchvision.io.read_video' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: 'torchvision.io.read_video(video_path, at the line number 407

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'torchvision.io.read_video(video_path,' to 'Proper Apostrophes Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: 'pts_unit='sec')', at the line number 407

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'pts_unit='sec')' to 'Proper Apostrophes Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: 'torch.zeros(self.num_frames, at the line number 410

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'torch.zeros(self.num_frames,' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: 'self.target_size,', at the line number 410

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'self.target_size,' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: 'self.target_size)', at the line number 410

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'self.target_size)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: 'real', at the line number 413

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'real' to 'Proper Hyphens Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: 'fake', at the line number 413

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'fake' to 'Proper Hyphens Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: '__init__', at the line number 413

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '__init__' to 'Proper Hyphens Mark.'.

Example: [view](#)

Part of the sentence to be reviewed: 'VideoViTEncoder', at the line number 417

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'VideoViTEncoder' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: 'fExpected', at the line number 418

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'fExpected' to 'Proper Quotation Mark.'

Example: [view](#)

Part of the sentence to be reviewed: [FakeAVCeleb](#), at the line number 436

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'FakeAVCeleb' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: [TensorBoard](#), at the line number 440

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'TensorBoard' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: [line up](#), at the line number 448

Category: **Compound Words**

Suggestion: Based on the context of the sentence, we recommend you to replace 'line up' to 'lineup'.

Example: [view](#)

Part of the sentence to be reviewed: [outputs/loss_curve.png](#), at the line number 452

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'outputs/loss_curve.png' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: [ssl_pretrain.py](#), at the line number 452

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'ssl_pretrain.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: [FakeAVCeleb](#), at the line number 454

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'FakeAVCeleb' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: [\(metrics/training_metrics_dashboard.png](#), at the line number 469

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '(metrics/training_metrics_dashboard.png' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: [finetune_deepfake.py.](#), at the line number 469

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'finetune_deepfake.py,' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: [FakeAVCeleb](#), at the line number 475

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'FakeAVCeleb' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[VoxCeleb](#)', at the line number 476

Category: **Orthographic Error**

Suggestion: Based on the context of the sentence, we recommend you to replace '[VoxCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[know how](#)', at the line number 476

Category: **Compound Words**

Suggestion: Based on the context of the sentence, we recommend you to replace '[know how](#)' to '[know-how](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[FakeAVCeleb](#)', at the line number 486

Category: **Orthographic Error**

Suggestion: Based on the context of the sentence, we recommend you to replace '[FakeAVCeleb](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[NeurIPS](#)', at the line number 497

Category: **Orthographic Error**

Suggestion: Based on the context of the sentence, we recommend you to replace '[NeurIPS](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '<https://github.com/yourusername/DeepDetect2>', at the line number 509

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '<https://github.com/yourusername/DeepDetect2>' to '[Proper Colon](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[DeepDetect2/](#)', at the line number 511

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[DeepDetect2/](#)' to '[Proper Slash Mark](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 511

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to '[Proper Slash Mark](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[Models/](#)', at the line number 511

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[Models/](#)' to '[Proper Slash Mark](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 512

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to '[Proper Fullstop/Period](#)'.

Example: [view](#)

Part of the sentence to be reviewed: '[Visual_encoder.py](#)', at the line number 512

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[Visual_encoder.py](#)' to '[Proper Fullstop/Period](#)'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 513

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['audio_encoder.py'](#), at the line number 513

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'audio_encoder.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['tokenizer'](#), at the line number 513

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace 'tokenizer' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 514

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['Fusion_Engine.py'](#), at the line number 514

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'Fusion_Engine.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 515

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['ssl_tasks.py'](#), at the line number 515

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'ssl_tasks.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 516

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: ['utils/'](#), at the line number 516

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'utils/' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 517

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[visual_preprocessing.py](#)', at the line number 517

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[visual_preprocessing.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 518

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[audio_preprocessing.py](#)', at the line number 518

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[audio_preprocessing.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 519

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Hyphens Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 520

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[dataset](#)', at the line number 520

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[dataset](#)' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 521

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[pretrain_datset_download.py](#)', at the line number 521

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[pretrain_datset_download.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 522

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[pretrain_prepare_dataset.py](#)', at the line number 522

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[pretrain_prepare_dataset.py](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 523

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['preprocess_dataset.py'](#), at the line number 523

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'preprocess_dataset.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 524

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['download_finetune_dataset.py'](#), at the line number 524

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'download_finetune_dataset.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 525

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['prepare_finetune_dataset.py'](#), at the line number 525

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'prepare_finetune_dataset.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 526

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['ssl_pretrain.py'](#), at the line number 526

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'ssl_pretrain.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['++'](#), at the line number 527

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '++' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: ['finetune_deepfake.py'](#), at the line number 527

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace 'finetune_deepfake.py' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 528

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[outputs/](#)', at the line number 528

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[outputs/](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[ssl_encoders_fused.pth](#)', at the line number 528

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[ssl_encoders_fused.pth](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[loss_curve.png](#)', at the line number 528

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[loss_curve.png](#)' to 'Proper Fullstop/Period'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 530

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[outputs_finetune/](#)', at the line number 530

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[outputs_finetune/](#)' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 531

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[metrics/](#)', at the line number 531

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[metrics/](#)' to 'Proper Slash Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[+--](#)', at the line number 532

Category: **Punctuation**

Suggestion: Based on the context of the sentence, we recommend you to replace '[+--](#)' to 'Proper Hyphens Mark'.

Example: [view](#)

Part of the sentence to be reviewed: '[TorchVision](#)', at the line number 536

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[TorchVision](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[TorchAudio](#)', at the line number 536

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[TorchAudio](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[tqdm](#)', at the line number 536

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[tqdm](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[tensorboard](#)', at the line number 536

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[tensorboard](#)' to Proper word.

Example: [view](#)

Part of the sentence to be reviewed: '[gdown](#)', at the line number 536

Category: **Orthographic Error**

Based on the context of the sentence, we recommend you to replace '[gdown](#)' to Proper word.

Example: [view](#)

Category Description:

What is Orthographic Error?

Orthographic errors occur when fail to understand the relationship between graphemes and phonemes.

For example, the authors will write skool instead of school or kik instead of kick. [\[top\]](#)

What is Punctuation Error?

Punctuational error often centre around misplacing punctuation in a sentence, incorrectly punctuating plural words, overusing and confusing the uses of different punctuation marks.

For example, Incorrect: I bought some olives, which we didn't eat when I went shopping last week.

Correct: 'I bought some olives, which we didn't eat, when I went shopping last week.' [\[top\]](#)

What is Compound Words?

Compound words occur when two or more words combine to form one individual word or a phrase that acts as one individual word. Compound words often produce writing mistakes because it's easy to forget if they're spelled as one word or two words.

For example, Incorrect: We ate icecream after the foot ball game at the local high school.

Correct: We ate ice cream after the football game at the local high school. [\[top\]](#)